

Kneo Agent Platform

Reference

Version 0.11.0

2026-06-15

*CLI usage and command reference, HTTP API contract,
environment-variable surface, examples walkthrough, and
project configuration — one printable reference.*

Table of contents

Table of contents	2
CLI usage	6
Common commands	6
Talking to a service	6
Profiles	7
Retry, timeout, and idempotency	7
Spec migration	8
Spec linting	8
Policy reports	9
Spec bundles	9
Kneo Serv CLI Reference	10
kneo	10
kneo config	10
kneo config show	11
kneo config init	11
kneo config resolve	11
kneo config secrets	11
kneo config render-spec	12
kneo config profile	12
kneo config profile set	12
kneo config profile use	13
kneo config profile list	13
kneo config profile show	13
kneo config profile delete	14
kneo spec	14
kneo spec validate	14
kneo spec compile	15
kneo spec resolve	15
kneo spec migrate	15
kneo spec policy-report	16
kneo spec bundle	16
kneo spec bundle sign	16
kneo spec bundle verify	17
kneo run	17
kneo runs	18
kneo runs get	18
kneo runs trace	18

kneo runs checkpoints	19
kneo runs replay	19
kneo runs checkpoint-diff	19
kneo runs cancel	20
kneo human	20
kneo human get	20
kneo human list	20
kneo human resume	21
kneo service	21
kneo service serve	21
Platform service API	23
Versioning	23
Authentication	23
Redaction	24
Spec governance diagnostics	24
Workflow specs	25
Declarative tools, MCP servers, and composition	26
Secret management	27
Environment policy management	28
Request limits	29
Structured logging	29
SDK OpenTelemetry tracing	29
Idempotency	30
Run cancellation	30
Retry, timeout, and backoff	30
Health checks	31
Background worker queue	31
Recovery and continuation	31
Replay and checkpoint diff	32
Audit events	32
Error responses	33
SQLite migrations	33
Retention and pruning	33
Checkpoint payload limits	34
Backup and restore	34
Runs	34
Human tasks	35
Specs	35
Skills	36
Audit	36

Security and policies	36
Worked examples	37
Health	37
Create a run	37
Get run state	38
List runs (paginated)	39
Cancel a run	39
Validate a spec	39
List human tasks	40
Resume a human task	41
List audit events	41
Inspect credential references	42
Read or update environment policy	42
Error response shape	43
Pagination, filtering, and sorting	43
Run lifecycle & failure semantics	
Status reference	45
Step failure: on_error semantics	46
Timeouts, cancellation, pause/resume	47
Async run creation returns 202 Accepted	47
What retention prunes	47
Error taxonomy quick reference	48
Environment variables	50
Project	50
Service auth	50
Persistence	51
Service limits	51
Run queue and workers	52
Runtime reliability	53
CLI client	54
Observability	54
Retention	56
Checkpoint storage	58
Examples	59
Feature → example matrix	59
Spec files	60
research_agent.yaml	60
graph_review_workflow.yaml	61
concurrent_review_workflow.yaml	61
group_chat_workflow.yaml	61

human_approval_workflow.yaml	62
run_with_timeout.py	63
smoke_human_workflow.yaml	63
declarative_spec.yaml	63
Project config	64
project_config.yaml	64
Helper Python	64
app_functions.py	64
human_functions.py	64
Adapting an example	64
0.9.0 additions: resilience, handoff, stdio MCP	65
0.10.0 additions: regression-showcase examples	65
Project .kneo/ config	67
Retention	68
Local / self-hosted LLM endpoints	68

CLI usage

Source: [docs/user/cli.md](#)

This page shows the day-to-day CLI patterns. The full subcommand and flag reference is generated from the Typer app and committed at [cli_reference.md](#); refresh it with:

```
python docs/script/generate_reference_docs.py
```

For first-time setup, see [quickstart.md](#).

Common commands

```
kneo config init --name research-agent-demo
kneo config show
kneo spec validate examples/research_agent.yaml
kneo spec lint examples/research_agent.yaml
kneo spec compile examples/research_agent.yaml
kneo spec migrate legacy_agent.yaml --output migrated_agent.yaml
kneo spec policy-report examples/research_agent.yaml --json
kneo spec validate examples/research_agent.yaml --env prod
kneo spec bundle sign examples/research_agent.yaml \
  --output bundles/research_agent.json \
  --approved-by release-manager --env prod
kneo spec bundle verify bundles/research_agent.json
kneo run examples/research_agent.yaml --input "Analyze Nvidia AI business"
kneo runs get <run_id>
kneo runs cancel <run_id>
kneo runs trace <run_id>
kneo runs checkpoints <run_id>
kneo runs replay <run_id>
kneo runs checkpoint-diff <run_id> --from-sequence 1 --to-sequence 3
kneo human get <continuation_id>
kneo human resume <continuation_id> --request-id <request_id> --approve
```

Talking to a service

For one-off service-backed calls, set environment variables:

```
export KNEO_SERV_API_KEY=<api-key>
kneo run examples/research_agent.yaml \
```

```
--service-url http://127.0.0.1:8000 \
--input "hello"
```

Profiles

For multiple service environments, store named profiles instead of re-typing URLs and tokens:

```
kneo config profile set local \
  --service-url http://127.0.0.1:8000 --api-key <api-key>
kneo config profile set staging \
  --service-url https://staging.example.com --api-key <api-key> --no-activate
kneo config profile list
kneo config profile use staging

kneo spec validate examples/research_agent.yaml --profile staging
kneo run examples/research_agent.yaml --profile staging --input "hello"
kneo runs get <run_id> --profile staging
kneo human list --profile staging
```

Profiles live at `~/.kneo_serv/profiles.json` with owner-only file permissions. Set `KNEO_SERV_PROFILES_PATH` to use a different location (handy in tests). The CLI reports whether a profile is authenticated but never prints stored API keys.

Service-connection precedence is:

1. `--service-url` (plus the API key from `--profile`, when given).
2. An explicit `--profile`.
3. `service.default_url` from `.kneo/config.yaml`.
4. The currently active CLI profile.
5. Local in-process execution.

Retry, timeout, and idempotency

Service-backed CLI commands honor retry and timeout settings:

```
export KNEO_SERV_CLIENT_TIMEOUT=120
export KNEO_SERV_CLIENT_RETRIES=2
export KNEO_SERV_CLIENT_RETRY_BACKOFF_SECONDS=0.25
```

Transient failures (408, 429, 5xx, plus network errors) are retried. Authentication, authorization, validation, and not-found failures fail fast with a clear message.

For retry-safe POSTs, set a stable idempotency key:

```
export KNEO_SERV_IDEMPOTENCY_KEY=<stable-client-generated-key>
```

The service replays the original response for duplicate `POST /runs` or human-task resume requests that carry the same key and payload.

Spec migration

Use `kneo spec migrate` to convert older or unversioned YAML specs to the current `version: v1` shape:

```
kneo spec migrate legacy_agent.yaml --output migrated_agent.yaml
kneo spec migrate migrated_agent.yaml --check --json
```

Migration currently supports unversioned specs, `v0`, and `v1`. Unsupported future versions fail fast rather than being guessed.

Spec linting

`kneo spec lint` runs the same validator pipeline as `kneo spec validate` but **filters output to warnings and errors only** (info-severity diagnostics are dropped) and **exits non-zero if any are found**. Use it as a CI gate to catch deprecated fields, unsafe imports, shorthand tool selection without explicit permissions, network/shell/ filesystem-write capabilities without an allow-list, and missing human approvals on privileged surfaces — without having to filter `spec validate` output by hand.

```
# CI gate – fails the build if the spec carries any warnings or errors
kneo spec lint examples/research_agent.yaml
echo "exit=$?"

# Machine-readable output with summary counts
kneo spec lint examples/research_agent.yaml --json
```

The JSON envelope is:

```
{
  "clean": false,
  "warning_count": 3,
  "error_count": 0,
  "diagnostics": [
    {
      "severity": "warning",
      "code": "W_TOOL_POLICY_UNRESTRICTED",
      "path": "...",
      "message": "...",
      "suggestion": "..."
    }
  ]
}
```



```
]
}
```

Lint is intentionally stricter than `validate` (which prints diagnostics but does not exit non-zero on warnings). For the full diagnostic list including info-severity items, use `kneo spec validate` instead.

Policy reports

`kneo spec policy-report` emits a structured governance report covering memory, tool permissions, guardrails, MCP imports, and human-review requirements:

```
kneo spec policy-report examples/research_agent.yaml --json
kneo spec policy-report examples/research_agent.yaml \
  --profile staging --json
```

The report includes validation diagnostics, so CI can block on errors while still surfacing warnings such as unrestricted network tools or missing human approval steps.

When project config declares `environments.<name>.policy_enforcement`, spec commands and `run --env <name>` enforce the selected environment after overlays and defaults are applied.

Spec bundles

`kneo spec bundle sign` produces an approved deployment bundle with a canonical spec digest and an HMAC signature:

```
export KNEO_SERV_SPEC_SIGNING_KEY=<deployment-signing-key>
kneo spec bundle sign examples/research_agent.yaml \
  --output bundles/research_agent.prod.json \
  --approved-by release-manager \
  --env prod
kneo spec bundle verify bundles/research_agent.prod.json
```

Verification fails if the bundle contents are edited after signing, or if a different signing key is used.

Kneo Serv CLI Reference

Source: docs/user/cli_reference.md

Generated by `python docs/script/generate_reference_docs.py`.

kneo

Usage: kneo [OPTIONS] COMMAND [ARGS]...

Kneo Agent Platform CLI

Options

<code>--version</code>	<code>-V</code>	Show the kneo-serv version and exit.
<code>--install-completion</code>		Install completion for the current shell.
<code>--show-completion</code>		Show completion for the current shell, to copy it or customize the installation.
<code>--help</code>		Show this message and exit.

Commands

<code>config</code>	Manage project config
<code>spec</code>	Validate and compile specs
<code>run</code>	Run agent/workflow specs
<code>runs</code>	Inspect runs
<code>human</code>	Manage human-in-the-loop tasks
<code>service</code>	Run or manage the Kneo service

kneo config

Usage: kneo config [OPTIONS] COMMAND [ARGS]...

Manage project config

Options

<code>--help</code>	Show this message and exit.
---------------------	-----------------------------

Commands

<code>show</code>	Show project config.
<code>init</code>	Create a <code>.kneo/config.yaml</code> project config.
<code>resolve</code>	Resolve project context, including environment and overlays.
<code>secrets</code>	Show configured secret references without exposing values.
<code>render-spec</code>	Render default spec plus environment overlays into one YAML

```

file.
profile    Manage service connection profiles

```

kneo config show

Usage: kneo config show [OPTIONS]

Show project config.

Options

```

--json
--help    Show this message and exit.

```

kneo config init

Usage: kneo config init [OPTIONS]

Create a .kneo/config.yaml project config.

Options

```

--name      TEXT [default: kneo-serv-project]
--force
--help      Show this message and exit.

```

kneo config resolve

Usage: kneo config resolve [OPTIONS]

Resolve project context, including environment and overlays.

Options

```

--env      TEXT
--spec     PATH
--json
--help      Show this message and exit.

```

kneo config secrets

Usage: kneo config secrets [OPTIONS]

Show configured secret references without exposing values.

Options

--env TEXT
 --json
 --help Show this message and exit.

kneo config render-spec

Usage: kneo config render-spec [OPTIONS]

Render default spec plus environment overlays into one YAML file.

Options

* --output -o PATH [required]
 --env TEXT
 --spec PATH
 --help Show this message and exit.

kneo config profile

Usage: kneo config profile [OPTIONS] COMMAND [ARGS]...

Manage service connection profiles

Options

--help Show this message and exit.

Commands

set Create or update a named service profile.
 use Set the active service profile.
 list List configured service profiles without exposing API keys.
 show Show one service profile without exposing its API key.
 delete Delete a service profile.

kneo config profile set

Usage: kneo config profile set [OPTIONS] NAME

Create or update a named service profile.

Arguments

* name TEXT [required]

Options

* --service-url TEXT [required]
 --api-key TEXT
 --activate --no-activate [default: activate]
 --json
 --help Show this message and exit.

kneo config profile use

Usage: kneo config profile use [OPTIONS] NAME

Set the active service profile.

Arguments

* name TEXT [required]

Options

--json
 --help Show this message and exit.

kneo config profile list

Usage: kneo config profile list [OPTIONS]

List configured service profiles without exposing API keys.

Options

--json
 --help Show this message and exit.

kneo config profile show

Usage: kneo config profile show [OPTIONS]

Show one service profile without exposing its API key.

Options

```
--name      TEXT
--json
--help      Show this message and exit.
```

kneo config profile delete

Usage: kneo config profile delete [OPTIONS] NAME

Delete a service profile.

Arguments

```
*   name      TEXT  [required]
```

Options

```
--help      Show this message and exit.
```

kneo spec

Usage: kneo spec [OPTIONS] COMMAND [ARGS]...

Validate and compile specs

Options

```
--help      Show this message and exit.
```

Commands

validate

```
lint      Surface validator warnings and errors. Exits 1 if any are
           found.
```

compile

resolve

```
migrate    Migrate an older YAML spec to the current v1 shape.
```

```
policy-report Generate a policy evaluation report for a spec.
```

```
bundle     Sign and verify approved spec bundles
```

kneo spec validate

Usage: kneo spec validate [OPTIONS] [SPEC_PATH]

Arguments

```
spec_path  [SPEC_PATH]
```

Options

<code>--env</code>	TEXT	
<code>--json</code>		
<code>--service-url</code>	TEXT	
<code>--profile</code>	TEXT	Service profile name
<code>--help</code>		Show this message and exit.

kneo spec compile

Usage: `kneo spec compile [OPTIONS] [SPEC_PATH]`

Arguments

<code>spec_path</code>	[SPEC_PATH]
------------------------	-------------

Options

<code>--env</code>	TEXT	
<code>--json</code>		
<code>--service-url</code>	TEXT	
<code>--profile</code>	TEXT	Service profile name
<code>--help</code>		Show this message and exit.

kneo spec resolve

Usage: `kneo spec resolve [OPTIONS] [SPEC_PATH]`

Arguments

<code>spec_path</code>	[SPEC_PATH]
------------------------	-------------

Options

<code>* --output -o</code>	PATH	[required]
<code>--env</code>	TEXT	
<code>--help</code>		Show this message and exit.

kneo spec migrate

Usage: `kneo spec migrate [OPTIONS] SPEC_PATH`

Migrate an older YAML spec to the current v1 shape.

Arguments

<code>* spec_path</code>	PATH	[required]
--------------------------	------	------------

```
Options
--output -o      PATH
--check
--json
--help          Show this message and exit.
```

kneo spec policy-report

Usage: kneo spec policy-report [OPTIONS] [SPEC_PATH]

Generate a policy evaluation report for a spec.

Arguments

```
spec_path      [SPEC_PATH]
```

Options

```
--env          TEXT
--json
--service-url  TEXT
--profile      TEXT  Service profile name
--help        Show this message and exit.
```

kneo spec bundle

Usage: kneo spec bundle [OPTIONS] COMMAND [ARGS]...

Sign and verify approved spec bundles

Options

```
--help        Show this message and exit.
```

Commands

```
sign  Create an approved, signed spec bundle.
verify Verify an approved spec bundle digest and signature.
```

kneo spec bundle sign

Usage: kneo spec bundle sign [OPTIONS] SPEC_PATH

Create an approved, signed spec bundle.

Arguments


```

*   spec_path      PATH  [required]

Options
* --output      -o      PATH  [required]
* --approved-by      TEXT  [required]
  --env          TEXT
  --key-env       TEXT  [default: KNEO_SERV_SPEC_SIGNING_KEY]
  --json
  --help                  Show this message and exit.

```

kneo spec bundle verify

Usage: kneo spec bundle verify [OPTIONS] BUNDLE_PATH

Verify an approved spec bundle digest and signature.

Arguments

```

*   bundle_path    PATH  [required]

```

Options

```

--key-env      TEXT  [default: KNEO_SERV_SPEC_SIGNING_KEY]
--json
--help                  Show this message and exit.

```

kneo run

Usage: kneo run [OPTIONS] [SPEC_PATH] COMMAND [ARGS]...

Run agent/workflow specs

Arguments

```

spec_path      [SPEC_PATH]  Path to Kneo YAML spec

```

Options

```

* --input      -i      TEXT  Input text [required]
  --target      TEXT    agent or workflow [default: workflow]
  --env          TEXT
  --json
  --service-url  TEXT
  --profile      TEXT    Service profile name
  --help                  Show this message and exit.

```

kneo runs

Usage: kneo runs [OPTIONS] COMMAND [ARGS]...

Inspect runs

Options

--help Show this message and exit.

Commands

get

trace

checkpoints

replay

checkpoint-diff

cancel

kneo runs get

Usage: kneo runs get [OPTIONS] RUN_ID

Arguments

* run_id TEXT [required]

Options

--json

--service-url TEXT

--profile TEXT Service profile name

--help Show this message and exit.

kneo runs trace

Usage: kneo runs trace [OPTIONS] RUN_ID

Arguments

* run_id TEXT [required]

Options

--json

--service-url TEXT

--profile TEXT Service profile name

--help Show this message and exit.

kneo runs checkpoints

Usage: kneo runs checkpoints [OPTIONS] RUN_ID

Arguments

* run_id TEXT [required]

Options

--json
 --service-url TEXT
 --profile TEXT Service profile name
 --help Show this message and exit.

kneo runs replay

Usage: kneo runs replay [OPTIONS] RUN_ID

Arguments

* run_id TEXT [required]

Options

--json
 --service-url TEXT
 --profile TEXT Service profile name
 --help Show this message and exit.

kneo runs checkpoint-diff

Usage: kneo runs checkpoint-diff [OPTIONS] RUN_ID

Arguments

* run_id TEXT [required]

Options

--from-sequence INTEGER
 --to-sequence INTEGER
 --json
 --service-url TEXT
 --profile TEXT Service profile name
 --help Show this message and exit.

kneo runs cancel

Usage: kneo runs cancel [OPTIONS] RUN_ID

Arguments

* run_id TEXT [required]

Options

--json
--service-url TEXT
--profile TEXT Service profile name
--help Show this message and exit.

kneo human

Usage: kneo human [OPTIONS] COMMAND [ARGS]...

Manage human-in-the-loop tasks

Options

--help Show this message and exit.

Commands

get
list
resume

kneo human get

Usage: kneo human get [OPTIONS] CONTINUATION_ID

Arguments

* continuation_id TEXT [required]

Options

--json
--service-url TEXT
--profile TEXT Service profile name
--help Show this message and exit.

kneo human list

Usage: kneo human list [OPTIONS]

Options

```
--json
--service-url      TEXT
--profile          TEXT  Service profile name
--help             Show this message and exit.
```

kneo human resume

Usage: kneo human resume [OPTIONS] CONTINUATION_ID

Arguments

```
* continuation_id  TEXT  [required]
```

Options

```
* --request-id      TEXT  [required]
--approve
--reject
--edit              TEXT
--provide           TEXT
--select            TEXT
--json
--service-url       TEXT
--profile           TEXT  Service profile name
--help             Show this message and exit.
```

kneo service

Usage: kneo service [OPTIONS] COMMAND [ARGS]...

Run or manage the Kneo service

Options

```
--help             Show this message and exit.
```

Commands

```
serve  Run the Kneo FastAPI service.
```

kneo service serve

Usage: kneo service serve [OPTIONS]

Run the Kneo FastAPI service.

Options

--host	TEXT	[default: 127.0.0.1]
--port	INTEGER	[default: 8000]
--reload		
--help		Show this message and exit.

Platform service API

Source: docs/user/service_api.md

The HTTP contract exposed by `kneo service serve`. The generated OpenAPI schema is committed at [openapi.json](#); refresh it with `python docs/script/generate_reference_docs.py`.

This page covers versioning, auth, redaction, governance, and the request / response shapes for each route group. The [Worked examples](#) section near the bottom has copy-pasteable `curl` invocations for the most common endpoints.

Versioning

The stable public HTTP API is exposed under `/v1`. Existing unversioned routes remain available for local development and backwards compatibility, but new service clients should prefer `/v1`.

Examples:

```
GET /v1/healthz
POST /v1/runs
GET /v1/runs/{run_id}
POST /v1/human-tasks/{continuation_id}/resume
```

Authentication

Authentication is disabled by default for local development. Enable it by setting API keys before starting the service:

```
export KNEO_SERV_AUTH_ENABLED=true
export KNEO_SERV_API_KEYS='operator:operator-token:operator;reviewer:reviewer-token:reviewer'
export KNEO_SERV_ADMIN_API_KEY='admin-token'
```

Clients authenticate with either header:

```
Authorization: Bearer <api-key>
X-Kneo-API-Key: <api-key>
```

Built-in roles:

- `admin`: all scopes
- `operator`: `runs:read`, `runs:write`, `specs:read`, `human:read`, `audit:read`, `credentials:read`, `policies:read`, `policies:write`

- `reviewer: runs:read, human:read, human:write, audit:read`
- `service: runs:read, runs:write, specs:read, human:read, human:write, audit:read, audit:write, credentials:read, policies:read`
- `viewer: runs:read, specs:read, human:read, audit:read`

`KNEO_SERV_API_KEYS` accepts semicolon-separated entries:

```
name:key:role_or_scope[,role_or_scope]
```

Example with explicit scopes:

```
export KNEO_SERV_API_KEYS='ci:ci-token:service;runs-reader:read-token:runs:read'
```

Redaction

Service responses, traces, checkpoints, and CLI JSON output are redacted before they are returned or persisted as checkpoints. Redaction covers common secret keys and inline values such as passwords, tokens, API keys, authorization headers, emails, and SSNs.

Spec governance diagnostics

Spec validation includes static governance diagnostics before deployment:

- Unsafe tool or function implementation imports such as direct `os`, `subprocess`, `shutil`, `socket`, `importlib`, or `builtins` primitives are reported as errors.
- Shorthand tool selection or missing tool permission policies are reported as warnings.
- Network tools without `allowed_domains`, shell-capable tools, and filesystem write access are reported as warnings.
- Specs that expose privileged tools or unsafe imports without a human workflow approval step receive a `W_HUMAN_APPROVAL_MISSING` warning.

These diagnostics are returned by `kneo spec validate`, `POST /specs/validate`, and strict compiler flows.

`POST /specs/policy-report` returns a structured policy report covering memory configuration, tool permissions, declared MCP imports, guardrail stages, human reviewers, and human approval requirements. Use it in deployment gates when a spec needs a machine-readable policy summary before signing or promotion.

`GET /runs/{run_id}/policy-report` returns the same shape but operates on the spec the run was started with — the service reads it out of the run's stored metadata, so operators auditing a deployed

run don't need to ship the bundle to the service themselves. Same `specs:read` scope as the spec-bundle route.

```
curl -H "Authorization: Bearer $KNEO_API_KEY" \
  https://kneo.example.com/v1/runs/run-7c2f.../policy-report
```

Returns `404` if the run id is unknown, `400` if the run carries no spec metadata (older runs from a pre-0.3.0 store), and `200` with `{"valid": <bool>, "report": {...}}` otherwise. Each call records a `spec.policy_reported` audit event scoped to the run id with `metadata.source = "run"`, so spec-bundle calls and run-keyed calls are distinguishable in the audit log.

Project-based CLI flows can enforce different gates per environment through `environments.<name>.policy_enforcement`. Enforcement runs after overlays and defaults are applied, so `dev`, `staging`, and `prod` can require progressively stricter tool permissions, human review, guardrails, or blocked diagnostic codes.

Redaction is a safety layer, not a replacement for secret management. Provider keys and credentials should still be supplied through deployment secret stores or environment variables rather than embedded in specs or request payloads.

Workflow specs

YAML specs can target SDK-backed workflow families while preserving service validation, tracing, cancellation, and run-result metadata:

- `sequential`: ordered `steps`.
- `graph`: keyed `nodes`, conditional `edges`, and a `start` node.
- `concurrent`: fan-out `participants` executed by the SDK concurrent workflow.
- `handoff`: `participants` plus a `selector`; `sequence` and `round_robin` selectors are supported.
- `group-chat` or `group_chat`: `participants` repeated for `rounds`.

Orchestration workflow participants use the same step shape as sequential workflow steps:

```
workflow:
  type: handoff
  name: review-handoff
  participants:
    - id: researcher
      kind: agent
      ref: research_agent
    - id: reviewer
      kind: agent
      ref: review_agent
```

```

selector:
  type: sequence
  sequence: [researcher, reviewer]

```

Participant ids must be unique, participant refs must resolve to declared components, handoff selector entries must reference participant ids, and group-chat `rounds` must be at least `1`.

Declarative tools, MCP servers, and composition

A spec wires tools and composes agents declaratively. A `tool` is backed by **exactly one** of three sources (the validator rejects zero or multiple with `E_TOOL_NO_BACKING` / `E_TOOL_MULTIPLE_BACKINGS`):

- `implementation`: a Python import path.
- `mcp`: a reference `{server: <name>, name?: <remote-tool>}` to a declared MCP server (the hybrid lazy-binding path — the connection is opened on first call, not at compile time).
- `agent`: the **agent-as-tool** pattern — names a `components.agents` entry, exposed to the parent agent as a tool. Its input schema is fixed to a single `input` string; author-declared `parameters` are ignored (`W_AGENT_TOOL_PARAMETERS_IGNORED`).

MCP servers are declared under a top-level `mcp_servers` block; each entry sets a `transport` (`stdio` needs `command`; `http` needs `url`; `sse` needs `sse_url`) plus optional TLS material (`verify`, `ca_bundle`, `client_cert`, `client_key` / `client_key_ref`). Setting `verify: false` disables TLS verification and emits `W_MCP_TLS_VERIFY_DISABLED`. Supply secrets by reference (`client_key_ref`) so they never land in persisted spec state.

An agent can itself be **backed by a workflow** (the workflow-as-agent pattern) via `as_agent: <workflow-name>`; only `name` / `description` / `system_prompt` are legal alongside `as_agent` (anything else is the workflow's job and is rejected with `E_AS_AGENT_FIELDS`).

```

mcp_servers:
  search:
    transport: http
    url: https://mcp.example.com/api
    client_key_ref: MCP_SEARCH_KEY
tools:
  web_search:
    description: Search the web via MCP.
    mcp: {server: search, name: search}
  ask_specialist:
    description: Delegate to the specialist agent.
    agent: specialist          # agent-as-tool
components:
  agents:
    specialist: {name: specialist}

```

```

pipeline_agent:
  name: pipeline_agent
  as_agent: review_pipeline  # workflow-as-agent
workflows:
  review_pipeline:
    type: sequential
    steps:
      - {id: s1, kind: agent, ref: specialist}

```

Components are built in dependency order (a topological sort over tool → agent → workflow references); a dependency cycle is rejected at validation with `E_BUILD_CYCLE`.

Secret management

`kneo_serv` resolves provider keys, MCP credentials, service tokens, and runtime-specific values through named environment-variable references. Project config stores only env-var names, never raw secret values:

```

secrets:
  provider_env:
    openai: OPENAI_API_KEY
  extra_env:
    mcp_default: MCP_API_KEY

```

The default provider mappings include `openai / openai-agents`, `anthropic`, `google`, and `google-adk`. The CLI can show a redacted inventory for deployment checks:

```
kneo config secrets --json
```

Native provider startup can fail fast when a required provider secret is missing:

```
export KNEO_SERV_REQUIRE_PROVIDER_SECRETS=true
```

Service API keys remain in `KNEO_SERV_API_KEY`, `KNEO_SERV_API_KEYS`, and `KNEO_SERV_ADMIN_API_KEY`; the secret inventory reports whether they are present without exposing values.

The service exposes the same redacted inventory for operators:

```
GET /v1/security/credentials?providers=openai&include_service_tokens=false
```

This endpoint requires `credentials:read`. The response reports configured provider, extra, and service-token references with `present` flags and redacted values:

```
{
  "inventory": {
    "providers": {
      "openai": {
        "name": "provider:openai",
        "env_var": "OPENAI_API_KEY",
        "present": true,
        "value": "[REDACTED]"
      }
    },
    "extra": {}
  }
}
```

Every successful credential inventory request records a `credential.inventory_accessed` audit event. Audit metadata includes counts and which reference names were present; raw secret values are never included.

Environment policy management

Environment policy enforcement can be managed through the service when a deployment needs operator-controlled gates outside checked-in project config:

```
GET /v1/policies/environment
GET /v1/policies/environment/prod
PUT /v1/policies/environment/prod
```

Reads require `policies:read`; writes require `policies:write`. A policy update stores validated `EnvironmentPolicyEnforcement` settings in the run state store's `project_metadata` table/key-value area:

```
{
  "enabled": true,
  "fail_on_warnings": false,
  "blocked_diagnostic_codes": [],
  "require_human_review": true,
  "require_tool_permissions": true,
  "deny_unrestricted_tools": true,
  "require_guardrails": false
}
```

The response includes the current policy and, for updates, the previous policy when one existed. Each successful update records a `policy.changed` audit event with the policy surface, environment, previous/current redacted policy payloads, and changed field names.

Request limits

The service rejects oversized request bodies before route handling and applies strict request-model validation for inline payloads. Unknown request fields are rejected with `422`, and bodies above the configured transport limit return `413`.

Default limits:

- `KNEO_SERV_MAX_BODY_BYTES` : 1048576
- `KNEO_SERV_MAX_INPUT_CHARS` : 20000
- `KNEO_SERV_MAX_HUMAN_CONTENT_CHARS` : 20000
- `KNEO_SERV_MAX_INLINE_SPEC_BYTES` : 262144
- `KNEO_SERV_MAX_OVERRIDES_BYTES` : 65536
- `KNEO_SERV_MAX_METADATA_BYTES` : 32768
- `KNEO_SERV_MAX_LIST_ITEMS` : 100
- `KNEO_SERV_MAX_PATH_CHARS` : 4096

Structured logging

API requests emit redacted JSON log records on the `kneo_serv.service` logger. Each request record includes `event=http_request`, `request_id`, `method`, `path`, `status code`, `duration`, `client IP` when available, and route-supplied `run`, `continuation`, or `trace IDs` when known.

Clients can send `X-Request-ID`; otherwise the service generates one. The response always includes the effective `X-Request-ID`.

Configuration:

- `KNEO_SERV_REQUEST_LOGS` : defaults to `true`
- `KNEO_SERV_LOG_LEVEL` : defaults to `INFO`

SDK OpenTelemetry tracing

When SDK telemetry support is installed, set `KNEO_SERV_OTEL_ENABLED=true` to attach `kneo_agent.observability.OpenTelemetryMiddleware` to SDK-backed agents. The middleware uses the OpenTelemetry global tracer provider, so exporters and resources can be configured with standard `OTEL_*` environment variables in the deployment environment.

Service defaults keep potentially sensitive span attributes disabled:

- `KNEO_SERV_OTEL_RECORD_ARGUMENTS` : defaults to `false`
- `KNEO_SERV_OTEL_RECORD_RESULTS` : defaults to `false`

Enable those only for trusted deployments where tool arguments and results are safe to emit to telemetry backends.

Idempotency

`POST /runs`, `POST /specs/run`, and `POST /human-tasks/{continuation_id}/resume` support the `Idempotency-Key` header. When the same key is reused with the same request payload, the service returns the original response without creating a duplicate run or submitting a second human decision.

```
Idempotency-Key: <stable-client-generated-key>
```

Reusing a key with a different payload returns `409` with `idempotency_key_conflict`.

The CLI service client can send a key per call in code, or read one from:

```
export KNEO_SERV_IDEMPOTENCY_KEY=<stable-client-generated-key>
```

Human-task resume also takes a store-backed continuation lock. If another process is already resuming the same continuation, the service returns `409` with `resource_locked`.

Run cancellation

`POST /runs/{run_id}/cancel` marks a pending or running run as `cancelled`. Background execution receives a cooperative cancellation token through the SDK run config `extra` payload, so service workflows, agents, runtimes, and wrapped workflow steps check cancellation before and after unit-of-work boundaries. A cancelled run is not overwritten as completed if execution returns after cancellation was requested.

Provider calls that do not expose an interrupt primitive can only stop at the next cooperative boundary after the provider returns.

Retry, timeout, and backoff

Service-client retries are configured with `KNEO_SERV_CLIENT_*` variables. Provider/runtime and MCP calls use the same conservative policy shape:

```
export KNEO_SERV_PROVIDER_RETRIES=2
export KNEO_SERV_PROVIDER_RETRY_BACKOFF_SECONDS=0.25
export KNEO_SERV_PROVIDER_TIMEOUT_SECONDS=120
```

```
export KNEO_SERV_MCP_RETRIES=2
export KNEO_SERV_MCP_RETRY_BACKOFF_SECONDS=0.25
export KNEO_SERV_MCP_TIMEOUT_SECONDS=30
```

Workflow steps can also set `on_error: retry`, `max_retries`, and `timeout_seconds` in YAML specs. Cancellation is never retried.

Health checks

This section is the API contract. For an on-call triage tree mapping each `/readyz` check to recovery actions, see [incident_response.md](#).

- `GET /healthz` : lightweight API health.
- `GET /livez` : process liveness.
- `GET /readyz` : readiness for API wiring, run state store, continuation store, durable run queue, runtime registry, tool registry, and configured provider or MCP secret dependencies.

Provider and MCP dependency checks are opt-in so local development does not fail when no real upstream credentials are configured:

```
export KNEO_SERV_HEALTH_PROVIDERS=openai,anthropic
export KNEO_SERV_HEALTH_MCP_SECRETS=mcp_default
```

If a configured readiness dependency is missing or unhealthy, `/readyz` returns `503` with a structured `not_ready` detail payload.

Background worker queue

Async run creation enqueues run IDs into the configured run state store before worker execution. SQLite and file stores persist queue records with status, attempt count, lease owner, lease expiry, and error details; in-memory stores keep the same contract for tests and local ephemeral use.

Workers claim queued or expired leased records, execute the run through the same `PlatformManager.execute_run` path, and then mark the queue record `completed` or `failed`. On service startup the default manager starts a worker so previously queued records can be resumed.

Recovery and continuation

Workflow execution stores live execution context on run state and persists step/node completion and failure checkpoints. For interrupted non-human sequential workflows, the service can report the completed steps, failed step, resume input, and next step index:

```
GET /runs/{run_id}/recovery
```

When `replay_context.can_continue` is true, the run can continue from the last completed step boundary:

```
POST /runs/{run_id}/continue
```

Graph workflows expose replay context from node checkpoints, but automatic continuation is limited to sequential workflows until graph edge state is persisted at each routing decision.

Replay and checkpoint diff

Operators can inspect a compact replay timeline without reading full checkpoint payloads:

```
GET /runs/{run_id}/replay
```

The response includes checkpoint sequence, type, step/node IDs, status, current execution position, pending human request ID, error summary, and the same replay context used by `/runs/{run_id}/recovery`.

Checkpoint diffs compare checkpoint state and metadata. By default the latest two checkpoints are compared:

```
GET /runs/{run_id}/checkpoints/diff
GET /runs/{run_id}/checkpoints/diff?from_sequence=1&to_sequence=3
```

The diff response reports added, removed, and changed flattened paths. Values are redacted before returning.

Audit events

The service records redacted audit events in the configured run state store for successful spec operations, run creation, run cancellation, run continuation, spec-run execution, and human-in-the-loop decisions.

```
GET /audit-events
GET /audit-events?event_type=run.created
GET /audit-events?run_id=<run_id>
GET /audit-events?limit=50&offset=50&sort_by=created_at&sort_order=desc
```


The audit list endpoint requires `audit:read` and returns events newest first. Each event includes `event_type`, `actor`, optional `run_id` and `continuation_id`, redacted metadata, and `created_at`. The response carries the same pagination metadata block as the other list endpoints — `count` (items on this page), `total`, `limit`, `offset`, `sort_by`, and `sort_order` (limit 1–1000, default 100; `sort_order` defaults to `desc`).

Error responses

Every 4xx/5xx response uses the envelope `{"detail": {"error": "<code>", "message": "<human-readable>", ...}}`, where `error` is a stable, snake_case machine code (e.g. `not_found`, `invalid_request`, `internal_error`, `queue_full`, `resource_locked`, `unauthorized`, `forbidden`) decoupled from internal exception names. Some errors carry extra context keys (e.g. `resource`, `queue_depth`, `required_scope`). 500 responses are opaque (`internal_error` with a generic message); the real cause is logged server-side, never returned to the client. These shapes are published in the OpenAPI schema as `ErrorResponse` / `ErrorDetail`.

SQLite migrations

SQLite state stores apply versioned migrations on startup. The migration table is `schema_migrations`, and the current schema covers run state, checkpoints, idempotency records, locks, durable run queue records, continuation records, audit event records, and project metadata records.

Existing unversioned SQLite databases are upgraded in place with `CREATE TABLE IF NOT EXISTS` and `CREATE INDEX IF NOT EXISTS` statements, so existing run payloads remain readable after migration.

Project metadata is used by service-managed environment policies. Upgrade coverage verifies that existing SQLite databases can create, persist, and reload policy metadata after migrations have applied.

Retention and pruning

`RetentionManager` provides an operator-callable pruning job for run state, checkpoints, completed or failed queue records, file-backed continuations, audit events, artifacts, and logs. It can be configured directly or through environment variables:

```
export KNEO_SERV_RETENTION_RUNS_DAYS=30
export KNEO_SERV_RETENTION_CHECKPOINTS_DAYS=30
export KNEO_SERV_RETENTION_QUEUE_DAYS=14
export KNEO_SERV_RETENTION_CONTINUATIONS_DAYS=30
export KNEO_SERV_RETENTION_AUDIT_DAYS=90
```

```
export KNEO_SERV_RETENTION_ARTIFACTS_DAYS=30
export KNEO_SERV_RETENTION_LOGS_DAYS=30
```

The platform manager exposes `prune_retention()` for embedded operators and future scheduled jobs.

Checkpoint payload limits

SQLite and file stores transparently compress large checkpoint payloads before writing them. If a checkpoint remains above the hard cap after compression, the store persists a bounded checkpoint preview that keeps run ID, checkpoint type, step/node IDs, timestamps, limited trace previews, and metadata describing the size reduction.

Defaults:

- `KNEO_SERV_CHECKPOINT_COMPRESS_BYTES` : 65536
- `KNEO_SERV_CHECKPOINT_MAX_BYTES` : 1048576
- `KNEO_SERV_CHECKPOINT_PREVIEW_CHARS` : 1200
- `KNEO_SERV_CHECKPOINT_MAX_LIST_ITEMS` : 20
- `KNEO_SERV_CHECKPOINT_MAX_DICT_ITEMS` : 50

Backup and restore

This section documents the Python backup API. For the operator-facing production procedure (PostgreSQL `pg_dump`, off-site rotation, restore verification, DR checklist), see [backup_and_recovery.md](#).

The default SQLite store can be backed up online with SQLite's backup API:

```
from kneo_serv.maintenance import backup_sqlite_database, restore_sqlite_database

backup_sqlite_database(".kneo/kneo_runs.sqlite", ".kneo/backups/kneo_runs.sqlite")
restore_sqlite_database(".kneo/backups/kneo_runs.sqlite", ".kneo/kneo_runs.restored.sqlite")
```

The smoke test covers run state and checkpoint restore from the copied database. File-backed continuations, artifacts, and logs should be included in deployment-level filesystem backups when those paths are used.

Runs

- POST `/v1/runs`
- GET `/v1/runs`

- GET /v1/runs/{run_id}
- POST /v1/runs/{run_id}/cancel
- GET /v1/runs/{run_id}/policy-report
- GET /v1/runs/{run_id}/recovery
- GET /v1/runs/{run_id}/replay
- POST /v1/runs/{run_id}/continue
- GET /v1/runs/{run_id}/checkpoints
- GET /v1/runs/{run_id}/checkpoints/diff
- GET /v1/runs/{run_id}/trace

Legacy aliases:

- GET /runs
- POST /runs
- GET /runs/{run_id}
- POST /runs/{run_id}/cancel
- GET /runs/{run_id}/recovery
- GET /runs/{run_id}/replay
- POST /runs/{run_id}/continue
- GET /runs/{run_id}/checkpoints
- GET /runs/{run_id}/checkpoints/diff
- GET /runs/{run_id}/trace

Human tasks

- GET /v1/human-tasks
- GET /v1/human-tasks/{continuation_id}
- POST /v1/human-tasks/{continuation_id}/resume

Legacy aliases:

- GET /human-tasks
- GET /human-tasks/{continuation_id}
- POST /human-tasks/{continuation_id}/resume

Specs

- POST /v1/specs/validate
- POST /v1/specs/compile
- POST /v1/specs/explain
- POST /v1/specs/policy-report
- POST /v1/specs/run

All four read-only endpoints accept the same envelope (`spec_path` or inline `spec` , plus `environment` , `overlays` , and `overrides`); `compile` additionally honors `strict` . The overlays/overrides layer the effective spec the same way a run does. An invalid spec returns `400 spec_invalid` carrying the diagnostic list (it is not a `500`).

Legacy aliases:

- `POST /specs/validate`
- `POST /specs/compile`
- `POST /specs/explain`
- `POST /specs/policy-report`
- `POST /specs/run`

Skills

- `GET /v1/skills`

Read-only catalog of skills discovered in the service's default locations (`name` / `description` / `path`), paginated. Requires `specs:read` . No compilation or spec is needed. A run can toggle the root agent's skills per-request with the `skills` overlay on `POST /v1/runs` (`{add, disable}`); `add` may only enable skills already **declared** in the spec — a request cannot inject an undeclared skill.

Legacy alias:

- `GET /skills`

Audit

- `GET /v1/audit-events`

Legacy alias:

- `GET /audit-events`

Security and policies

- `GET /v1/security/credentials`
- `GET /v1/policies/environment`
- `GET /v1/policies/environment/{environment}`
- `PUT /v1/policies/environment/{environment}`

Legacy aliases:

- GET /security/credentials
- GET /policies/environment
- GET /policies/environment/{environment}
- PUT /policies/environment/{environment}

Worked examples

Concrete `curl` invocations and abbreviated response shapes for the most common endpoints. The full schema is in [openapi.json](#); these are illustrative.

All examples assume:

```
export BASE=http://127.0.0.1:8000
export KEY=operator-token # an entry from KNEO_SERV_API_KEYS
```

Health

```
curl -sf "$BASE/livez"      # {"ok": true, "metadata": {}}
curl -sf "$BASE/readyz"    # 200 with checks: {} or 503 with not_ready details
```

`/livez` and `/readyz` are intentionally unauthenticated. See [troubleshooting.md § 1.2](#) for the failure shape.

Create a run

Required scope: `runs:write`.

```
curl -sf -X POST "$BASE/v1/runs" \
-H "Authorization: Bearer $KEY" \
-H 'Content-Type: application/json' \
-d '{
  "input": "Summarize Nvidia AI strategy",
  "spec_path": "examples/research_agent.yaml",
  "target": "workflow",
  "environment": "prod",
  "async_mode": false
}' | jq
```

Synchronous response (run finished within the request):

```
{
  "run_id": "run_2026-05-10T12:34:56_a1b2c3d4",
```

```

"status": "succeeded",
"output_text": "Nvidia's AI strategy hinges on ...",
"human_intervention_required": false,
"continuation_id": null,
"metadata": {"workflow_kind": "sequential", "trace_event_count": 7}
}

```

If the workflow pauses on a human step:

```

{
  "run_id": "run_...",
  "status": "paused",
  "output_text": null,
  "human_intervention_required": true,
  "continuation_id": "cont_...",
  "metadata": {"pending_human_request": {"request_id": "req_...", "prompt": "Approve the
draft?"}}
}

```

For retry-safe submissions, send an `Idempotency-Key` header. Reusing the same key with the same body replays the original response; mismatched bodies return `409 idempotency_key_conflict`.

Get run state

Required scope: `runs:read`.

```

curl -sf "$BASE/v1/runs/run_..." \
  -H "Authorization: Bearer $KEY" | jq

```

```

{
  "run_id": "run_...",
  "status": "running",
  "agent_name": "research-copilot",
  "workflow_name": "research-pipeline",
  "workflow_kind": "sequential",
  "current_step_index": 1,
  "current_node_id": "analyze",
  "visited_steps": ["retrieve"],
  "visited_nodes": ["retrieve"],
  "trace_event_count": 4,
  "metadata": {"environment": "prod"}
}

```

For terminal status:

```
{
  "run_id": "run_...",
  "status": "succeeded",
  "output_text": "...",
  "visited_steps": ["retrieve", "analyze", "summarize"],
  "trace_event_count": 11
}
```

List runs (paginated)

```
curl -sf "$BASE/v1/runs?status=running&limit=20&sort_by=created_at&sort_order=desc" \
-H "Authorization: Bearer $KEY" | jq
```

```
{
  "runs": [
    {
      "run_id": "run_...",
      "status": "running",
      "workflow_name": "research-pipeline",
      "created_at": "2026-05-10T12:30:00Z"
    },
    {
      "run_id": "run_...",
      "status": "running",
      "workflow_name": "approval-workflow",
      "created_at": "2026-05-10T12:28:11Z"
    }
  ],
  "count": 2,
  "total": 2,
  "limit": 20,
  "offset": 0,
  "sort_by": "created_at",
  "sort_order": "desc"
}
```

Cancel a run

```
curl -sf -X POST "$BASE/v1/runs/run_.../cancel" \
-H "Authorization: Bearer $KEY"
```

The run transitions to `cancelled`; cancellation is cooperative — in-flight steps stop at unit-of-work boundaries. See [troubleshooting.md § 5.2](#).

Validate a spec

Required scope: `specs:read`.

```
curl -sf -X POST "$BASE/v1/specs/validate" \
-H "Authorization: Bearer $KEY" \
```

```
-H 'Content-Type: application/json' \
-d '{"spec_path": "examples/research_agent.yaml", "environment": "prod"}' | jq
```

```
{
  "valid": true,
  "diagnostics": [],
  "report": {
    "agent_name": "research-copilot",
    "workflow_name": "research-pipeline"
  }
}
```

For an invalid spec, `valid` is `false` and `diagnostics` is populated:

```
{
  "valid": false,
  "diagnostics": [
    {
      "severity": "error",
      "code": "E_UNKNOWN_TOOL",
      "message": "Tool 'web_search' is not registered.",
      "path": "agent.tools[0]"
    }
  ]
}
```

List human tasks

Required scope: `human:read` .

```
curl -sf "$BASE/v1/human-tasks?run_id=run_..." \
-H "Authorization: Bearer $KEY" | jq
```

```
{
  "tasks": [
    {
      "continuation_id": "cont_...",
      "run_id": "run_...",
      "request": {
        "request_id": "req_...",
        "prompt": "Approve the draft?",
        "deadline_epoch": 1715432400
      }
    }
  ],
}
```



```

"count": 1,
"total": 1,
"limit": 100,
"offset": 0
}

```

Resume a human task

Required scope: `human:write`. Pair with `Idempotency-Key` for safe retries.

```

curl -sf -X POST "$BASE/v1/human-tasks/cont_.../resume" \
-H "Authorization: Bearer $KEY" \
-H "Idempotency-Key: $(uuidgen)" \
-H 'Content-Type: application/json' \
-d '{
  "request_id": "req_...",
  "decision": "approved",
  "content": "Looks good. Ship it."
}' | jq

```

```

{
  "run_id": "run_...",
  "status": "succeeded",
  "output_text": "Published. https://...",
  "human_intervention_required": false,
  "continuation_id": null,
  "metadata": {}
}

```

`decision` is one of `approved`, `rejected`, `edited`, `selected`, `provided`. See [human_in_the_loop.md](#).

List audit events

Required scope: `audit:read`. Audit payloads are redacted; secret and PII patterns never appear.

```

curl -sf "$BASE/v1/audit-events?event_type=human.decision" \
-H "Authorization: Bearer $KEY" | jq

```

```

{
  "events": [
    {
      "id": "evt_...",
      "event_type": "human.decision",

```

```

    "actor": "reviewer",
    "occurred_at": "2026-05-10T12:35:01Z",
    "payload": {
      "request_id": "req_...",
      "decision": "approved",
      "selected_option": null,
      "result_status": "succeeded",
      "content_present": true
    }
  ],
  "count": 1
}

```

Inspect credential references

Required scope: `credentials:read`. Returns *presence* metadata only; secret values never appear.

```

curl -sf "$BASE/v1/security/credentials" \
-H "Authorization: Bearer $KEY" | jq

```

```

{
  "inventory": {
    "providers": [
      {
        "name": "openai",
        "env_var": "OPENAI_API_KEY",
        "present": true,
        "value":
        """***REDACTED***""",
      },
      {
        "name": "anthropic",
        "env_var": "ANTHROPIC_API_KEY",
        "present": false,
        "value": null
      }
    ],
    "mcp": [],
    "service": [
      {
        "name": "operator",
        "env_var": "KNEO_SERV_API_KEYS",
        "present": true,
        "value":
        """***REDACTED***""",
      }
    ]
  }
}

```

Each access records a `credential.inventory_accessed` audit event.

Read or update environment policy

Read requires `policies:read`; write requires `policies:write`.

```

curl -sf "$BASE/v1/policies/environment/prod" \
-H "Authorization: Bearer $KEY" | jq

```

```
{
  "environment": "prod",
  "policy": {
    "enabled": true,
    "fail_on_warnings": false,
    "blocked_diagnostic_codes": ["E_UNSAFE_TOOL_IMPORT"],
    "require_human_review": false,
    "require_tool_permissions": true,
    "deny_unrestricted_tools": true,
    "require_guardrails": false
  }
}
```

```
curl -sf -X PUT "$BASE/v1/policies/environment/prod" \
-H "Authorization: Bearer $KEY" \
-H 'Content-Type: application/json' \
-d '{
  "enabled": true,
  "require_tool_permissions": true,
  "deny_unrestricted_tools": true,
  "blocked_diagnostic_codes": ["E_UNSAFE_TOOL_IMPORT", "E_UNSAFE_FUNCTION_IMPORT"]
}' | jq
```

The response includes `previous_policy` so you can audit what changed. Each write records a `policy.changed` audit event.

Error response shape

All error paths use the same envelope:

```
{
  "error": "forbidden",
  "message": "Missing required scope: runs:write",
  "required_scope": "runs:write"
}
```

Common error codes: `unauthorized` (401), `forbidden` (403), `invalid_request` (400), `not_found` (404), `idempotency_key_conflict` (409), `payload_too_large` (413), `not_ready` (503). Errors map through [service/errors.py](#).

Pagination, filtering, and sorting

List-style endpoints return the original collection field plus pagination metadata:

```
{
  "count": 25,
  "total": 91,
  "window": 10000,
  "limit": 25,
  "offset": 50,
  "sort_by": "updated_at",
  "sort_order": "desc"
}
```

`total` is the store-wide count; `window` is the newest-rows window a single list request fetches and pages over. On deployments where `total > window`, rows older than the window are not reachable through the list endpoint (use retention pruning or direct store queries for archival access), and `sort_order=asc` orders *within* the window — it does not surface the oldest rows overall.

Supported query parameters:

- `GET /v1/runs`: `status`, `limit`, `offset`, `sort_by`, `sort_order`
- `GET /v1/runs/{run_id}/checkpoints`: `type`, `limit`, `offset`, `sort_by`, `sort_order`
- `GET /v1/runs/{run_id}/trace`: `event_type`, `limit`, `offset`, `sort_by`, `sort_order`
- `GET /v1/human-tasks`: `run_id`, `workflow_kind`, `status`, `limit`, `offset`, `sort_by`, `sort_order`

`sort_order` is `asc` or `desc`; `limit` is capped at 1000.

Only the documented query parameters above are honored. **0.11.0 breaking change:** an unrecognized query parameter is now **rejected with 422** (error: `"unknown_query_parameters"`, naming the offending keys) on the authenticated `/v1` (and root) routes — through 0.10.x it was silently ignored. This mirrors the request-body contract (unknown body fields are already rejected). The `/healthz`, `/readyz`, and `/metrics` endpoints are **exempt** (monitoring tooling may pass arbitrary scrape params). See [upgrade.md § 0.11.0](#).

Run lifecycle & failure semantics

Source: docs/user/run_lifecycle.md

The authoritative operator reference for what each run status means, how runs move between them, what happens when a step fails, times out, is cancelled, or pauses for a human — and which statuses retention prunes. The 0.9.0 cut made the state machine honest under failure; this page is the contract those fixes implement, and the surface the [contract-stability_policy](#) binds to for run semantics.

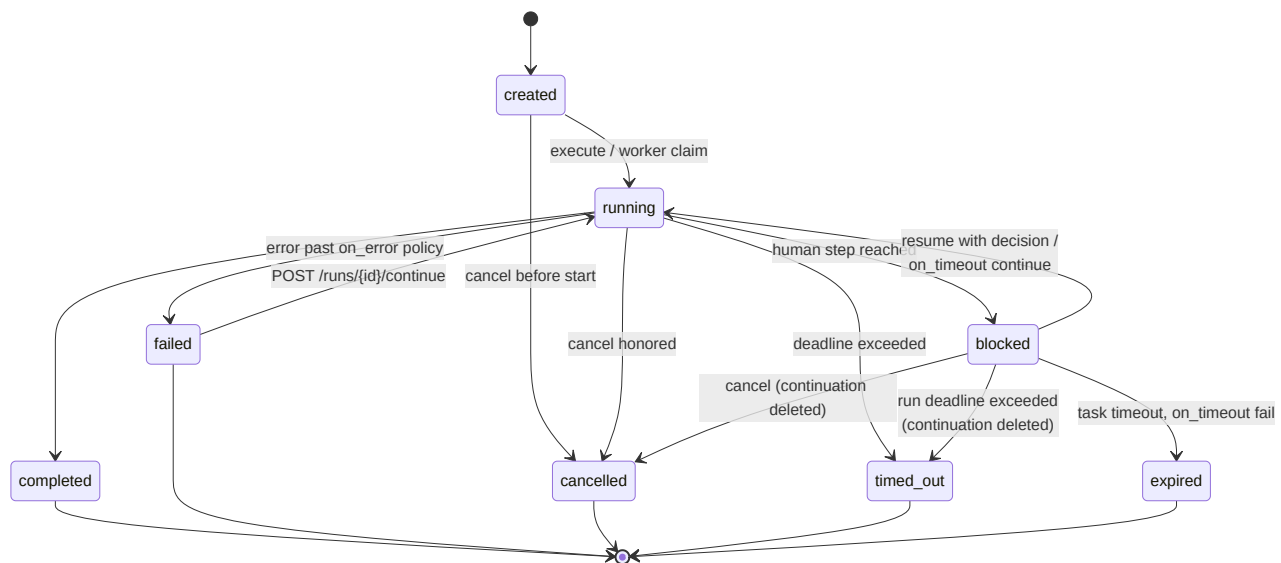
Related deep-dives: [human_in_the_loop.md](#) (the pause/resume walkthrough), [service_api.md](#) (routes and error envelopes), [troubleshooting.md](#).

Status reference

Status	Terminal?	Meaning
<code>created</code>	no	Run row persisted; not yet executing.
<code>queued</code> (<i>queue record</i>)	no	Waiting for a worker (async mode). The run row itself stays <code>created</code> / <code>running</code> ; <code>queued</code> is the queue record's status.
<code>running</code>	no	A worker (or the synchronous caller) is executing it under a lease.
<code>blocked</code>	no	Paused at a human step; a <code>WorkflowContinuation</code> holds the resume context.
<code>paused</code>	no	Reserved/legacy pause marker; human pauses use <code>blocked</code> .
<code>completed</code>	yes	Finished successfully.
<code>failed</code>	yes	A step or the runtime raised past its error policy. Eligible for <code>/continue</code> (sequential workflows).
<code>cancelled</code>	yes	An operator cancel landed (cooperatively honored mid-

Status	Terminal?	Meaning
		run).
<code>timed_out</code>	yes	The run exceeded its <code>timeout_seconds</code> deadline.
<code>expired</code>	yes	A blocked human task hit its timeout with <code>on_timeout: fail</code> .

Terminal statuses are final: every write path fences on the canonical terminal set, so a late worker result, a resume, or a cancel can never overwrite one (`409 run_state_conflict` where a client asks for it).



Step failure: `on_error` semantics

Each workflow step/node declares `on_error` (default `fail`). The policy applies **after** the step's retry budget (`max_retries`) is exhausted:

- **fail** — the run fails; the error lands in `state.error` and the `run_failed` checkpoint (with the collected trace).
- **retry** — the step retries up to `max_retries` before `fail` semantics apply.
- **continue** — the failed step is skipped: its **input passes through unchanged** to the next step, the failure is recorded in the step's checkpoint metadata (and a `step_failed` checkpoint on graph paths), and the run proceeds. In graph workflows, edge conditions see the real outcome — a `status_is: failed` condition routes the failure branch.
- **fallback** — the referenced step/node (`fallback_ref`) runs in place of the failed one, under the fallback's own retry/timeout policy but never its `on_error` policy (fallback chains

cannot recurse); traversal then continues. Note: a sequential `fallback_ref` is another step of the same workflow, so it also runs at its own chain position.

Cancellation and human-intervention pauses are **control flow, never step failures** — no `on_error` policy can swallow them. One unsupported corner: a `fallback_ref` pointing at a *human* step cannot pause — keep human steps in the main chain. `/specs/validate` rejects that shape (`E_STEP_FALLBACK_HUMAN`), along with a `fallback_ref` to a disabled graph node (`E_GRAPH_NODE_FALLBACK_DISABLED`).

Timeouts, cancellation, pause/resume

- **Run timeout** (`timeout_seconds`): the sweep marks the run `timed_out`, signals any still-executing worker to stop cooperatively, and deletes a blocked run's continuation.
- **Cancel** (`POST /runs/{id}/cancel`): cooperative mid-run; on a blocked run it also deletes the continuation (the task disappears from `GET /human-tasks`). Cancelling an already-terminal run is a no-op that returns the unchanged state.
- **Pause/resume**: a human step persists a continuation and the run goes `blocked`. Resume re-enters under the continuation lock and **refuses any run that is no longer `blocked`** with `409 run_state_conflict` — a cancel that raced the reviewer wins. The resumed leg runs under the run's cancellation token, and the full pre-pause trace survives on `GET /runs/{id}/trace`.
- **/continue** (crash/failure recovery, sequential only): re-enters a `failed` run from its last checkpoint. Terminal-but-not-failed runs are refused, as are `running` runs whose worker lease is still live (double-execution guard); an expired lease admits the legitimate crash-recovery case.

Async run creation returns `202 Accepted`

`POST /v1/runs` with `async_mode=true` returns **`202 Accepted`**: the run is dispatched to the worker queue and the caller polls `GET /runs/{id}` for progress. Synchronous creates (`async_mode=false`) return `200`.

0.11.0 breaking change: through 0.10.x this returned `200`; the `200 → 202` move was held for the contract-stability boundary and shipped in 0.11.0 (see [contract_stability.md](#) and [upgrade.md § 0.11.0](#)). The response body shape is unchanged — only the status code. (The run *row* progresses `created → running → terminal`; `queued` is the queue record's status, not the run's — see the status table above.)

What retention prunes

Each knob is days-based; `None` /unset means "never prune" (see [deployment.md](#) for configuration):

Knob	Prunes	Notes
<code>runs_days</code>	Terminal runs	Default statuses: <code>completed</code> , <code>failed</code> , <code>cancelled</code> , <code>timed_out</code> , <code>expired</code> ; override via <code>KNEO_SERV_RETENTION_RUN_STATUSES</code> .
<code>checkpoints_days</code>	Checkpoint rows	By <code>created_at</code> .
<code>queue_days</code>	Finished queue records	<code>completed</code> / <code>failed</code> queue statuses.
<code>continuations_days</code>	Stale continuations	Continuations of still- blocked runs are protected — age alone never bricks a pending human task.
<code>audit_days</code>	Audit events	By <code>created_at</code> .
<code>idempotency_days</code>	Idempotency records	By <code>updated_at</code> (a re-upserted key refreshes its lifetime).
<code>artifacts_days</code> / <code>logs_days</code>	Files under <code>KNEO_SERV_ARTIFACT_PATH</code> / <code>KNEO_SERV_LOG_PATH</code>	By mtime.

Error taxonomy quick reference

HTTP	error	When
400	<code>spec_invalid</code> / <code>invalid_request</code>	Spec fails compile; malformed input.
403	<code>environment_policy_blocked</code>	The environment's stored policy blocks the deployment (diagnostics included).
404	<code>not_found</code>	Unknown run/continuation id.
409	<code>run_state_conflict</code>	Resume/continue against a run whose status forbids it.

HTTP	error	When
409	<code>resource_locked</code> / <code>idempotency_key_conflict</code>	Concurrent operation holds the lock; same key, different payload.
413	payload too large	Body over the cap — including chunked uploads.
422	<code>guardrail_violation</code>	A guardrail blocked the content (violation type included).
422	<i>(native envelope)</i>	Request-shape validation errors.
503	<code>queue_full</code>	Backpressure: retry after the queue drains (<code>Retry-After</code>).
503	<code>store_unavailable</code>	The persistence backend is unreachable; transient (<code>Retry-After</code>).

Full envelopes and examples: [service_api.md](#).

Environment variables

Source: <docs/user/environment.md>

Every environment variable read by `kneo-serv`, grouped by surface area. Defaults shown are the values used when the variable is unset.

For deployment templates that exercise these in context, see tutorial/postgres_deployment.md and `deploy/*.env.example`. The full HTTP contract that uses these knobs lives in service_api.md.

Project

Variable	Default	Purpose
<code>KNEO_PROJECT_CONFIG</code>	auto-discovery	Explicit path to <code>.kneo/config.yaml</code> .
<code>KNEO_ENV</code>	<code>dev</code>	Default project environment when <code>--env</code> is not provided.
<code>KNEO_SERV_SPEC_SIGNING_KEY</code>	unset	HMAC signing key used by <code>kneo spec bundle sign</code> and <code>verify</code> .

Service auth

Variable	Default	Purpose
<code>KNEO_SERV_AUTH_ENABLED</code>	enabled when API keys are configured	Require API keys on protected HTTP routes.
<code>KNEO_SERV_API_KEYS</code>	empty	Semicolon-separated <code>name:key:role_or_scope[,role_or_scope]</code> entries.
<code>KNEO_SERV_ADMIN_API_KEY</code>	empty	Admin API key with all scopes.
<code>KNEO_SERV_API_KEY</code>	empty	Client API key used by <code>ServiceClient</code> and one-off CLI calls.

Variable	Default	Purpose
KNEO_SERV_IDEMPOTENCY_KEY	empty	Stable idempotency key for retry-safe service POST calls.

Persistence

Variable	Default	Purpose
KNEO_SERV_DATABASE_URL	empty	PostgreSQL DSN. When set, service stores use PostgreSQL instead of SQLite plus file continuations. Requires <code>kneo-serv[postgres]</code> or <code>kneo-serv[deploy]</code> .
KNEO_SERV_PROFILES_PATH	<code>~/.kneo_serv/profiles.json</code>	CLI service-profile store location.

Service limits

Variable	Default	Purpose
KNEO_SERV_MAX_BODY_BYTES	1048576	Maximum HTTP request body size.
KNEO_SERV_MAX_INPUT_CHARS	20000	Maximum run input size.
KNEO_SERV_MAX_HUMAN_CONTENT_CHARS	20000	Maximum human response content size.
KNEO_SERV_MAX_INLINE_SPEC_BYTES	262144	Maximum inline spec payload size.
KNEO_SERV_MAX_OVERRIDES_BYTES	65536	Maximum spec override payload size.
KNEO_SERV_MAX_METADATA_BYTES	32768	Maximum request metadata payload size.
KNEO_SERV_MAX_LIST_ITEMS	100	Maximum requested list page size.

Variable	Default	Purpose
KNEO_SERV_MAX_PATH_CHARS	4096	Maximum path field size.

Run queue and workers

Variable	Default	Purpose
KNEO_SERV_WORKER_CONCURRENCY	1	In-process worker threads draining the run queue. Raise for provider-bound workloads so runs overlap on provider I/O. See performance.md .
KNEO_SERV_WORKER_LEASE_SECONDS	300	Queue-lease liveness window per claimed run (not a run-time cap). A running worker renews its lease every ~ 1/3 of this interval (the lease heartbeat), so a healthy long run keeps its lease; only a dead or starved worker lets it expire, after which the run is re-claimed. Set it above a single step's wall-clock, not the whole run.
KNEO_SERV_QUEUE_MAX_ATTEMPTS	5	Dead-letter cap. A run re-leased more than this many times (e.g. one that repeatedly crashes its worker) is failed with a <code>dead_letter</code> reason instead of retried forever. 0 disables the cap.
KNEO_SERV_MAX_QUEUE_DEPTH	0	Overload backpressure: <code>POST /v1/runs (async)</code> returns 503 once this many runs are queued. 0 disables backpressure (unbounded queue).
KNEO_SERV_SHUTDOWN_TIMEOUT_SECONDS	30	How long SIGTERM shutdown waits for workers to finish their

Variable	Default	Purpose
		in-flight run. A run still executing past this is interrupted by process exit, then released/retried after <code>KNEO_SERV_WORKER_LEASE_SECONDS</code> (not lost). Set this — and the orchestrator grace period — $\geq$ your longest run step to drain without a restart.

Runtime reliability

Variable	Default	Purpose
<code>KNEO_SERV_TOKEN_BUDGET</code>	unset	Deployment-wide per-run token ceiling (input + output across a run). A run that crosses it is failed with a <code>token_budget_exceeded 4xx</code> . This is a post-run / boundary check, not a mid-flight hard stop — the run completes the in-flight step, then fails. A spec's <code>model.token_budget</code> overrides this default per agent; unset disables the ceiling.
<code>KNEO_SERV_PROVIDER_RETRIES</code>	0	Provider/runtime retry count.
<code>KNEO_SERV_PROVIDER_RETRY_BACKOFF_SECONDS</code>	0.0	Provider/runtime retry backoff (seconds between attempts).
<code>KNEO_SERV_PROVIDER_TIMEOUT_SECONDS</code>	unset	Provider/runtime timeout.
<code>KNEO_SERV_MCP_RETRIES</code>	0	MCP tool retry count.
<code>KNEO_SERV_MCP_RETRY_BACKOFF_SECONDS</code>	0.0	MCP retry backoff (seconds between attempts).

Variable	Default	Purpose
<code>KNEO_SERV_MCP_TIMEOUT_SECONDS</code>	<code>unset</code>	MCP tool timeout.
<code>KNEO_SERV_REQUIRE_PROVIDER_SECRETS</code>	<code>false</code>	Fail native provider setup when provider secrets are absent.
<code>KNEO_SERV_RUN_PROVIDER_INTEGRATION</code>	<code>false</code>	Enable opt-in real provider integration tests. Requires provider credentials such as <code>OPENAI_API_KEY</code> .
<code>KNEO_SERV_PROVIDER_TEST_MODEL</code>	<code>gpt-4o-mini</code>	OpenAI model used by the opt-in provider integration smoke test.
<code>KNEO_SERV_RUN_POSTGRES_INTEGRATION</code>	<code>false</code>	Enable opt-in PostgreSQL persistence smoke tests. Requires <code>KNEO_SERV_DATABASE_URL</code> and the <code>postgres</code> extra.

CLI client

Variable	Default	Purpose
<code>KNEO_SERV_CLIENT_TIMEOUT</code>	<code>120</code>	HTTP client timeout in seconds.
<code>KNEO_SERV_CLIENT_RETRIES</code>	<code>2</code>	Service-client retry count for transient failures.
<code>KNEO_SERV_CLIENT_RETRY_BACKOFF_SECONDS</code>	<code>0.25</code>	Service-client retry backoff.

Observability

Variable	Default	Purpose
<code>KNEO_SERV_REQUEST_LOGS</code>	<code>true</code>	Enable structured request logs.

Variable	Default	Purpose
<code>KNEO_SERV_LOG_LEVEL</code>	<code>INFO</code>	Logging level for the whole stack — the <code>kneo_serv.service</code> (request), <code>kneo_serv.platform</code> (worker / lease / drain), and <code>kneo_agent</code> SDK loggers are all set from this one value at startup.
<code>KNEO_SERV_AUDIT_EXPORT_ENABLED</code>	<code>false</code>	When <code>true</code> , every persisted (already-redacted) audit event is also emitted as a JSON line on the <code>kneo_serv.audit</code> logger — attach a handler to forward it to a file / syslog / SIEM. Off = zero behavior change. See observability.md § Audit-event export .
<code>KNEO_SERV_OTEL_ENABLED</code>	<code>false</code>	Attach <code>kneo_agent.observability.OpenTelemetryMiddleware</code> to SDK-backed agents. Requires <code>kneo-serv[telemetry]</code> or <code>kneo-serv[deploy]</code> .
<code>KNEO_SERV_OTEL_RECORD_ARGUMENTS</code>	<code>false</code>	Record tool-call arguments in SDK OpenTelemetry spans. Keep disabled when arguments may contain PII or secrets.
<code>KNEO_SERV_OTEL_RECORD_RESULTS</code>	<code>false</code>	Record tool results in SDK OpenTelemetry spans. Keep disabled for large or sensitive results.
<code>KNEO_SERV_HEALTH_PROVIDERS</code>	<code>empty</code>	Comma-separated provider secret names to include in readiness checks.
<code>KNEO_SERV_HEALTH_MCP_SECRETS</code>	<code>empty</code>	Comma-separated MCP secret names to include in readiness checks.

Variable	Default	Purpose
<code>KNEO_SERV_METRICS_ENABLED</code>	<code>true</code>	Mount the unauthenticated Prometheus <code>/metrics</code> endpoint. Restrict it to your monitoring network — see observability.md .

Retention

Variable	Default	Purpose
<code>KNEO_SERV_RETENTION_RUNS_DAYS</code>	<code>unset</code>	Delete old runs with terminal statuses.
<code>KNEO_SERV_RETENTION_CHECKPOINTS_DAYS</code>	<code>unset</code>	Delete old checkpoints.
<code>KNEO_SERV_RETENTION_QUEUE_DAYS</code>	<code>unset</code>	Delete old completed or failed queue records.
<code>KNEO_SERV_RETENTION_CONTINUATIONS_DAYS</code>	<code>unset</code>	Delete old continuations. Continuations of still- <code>blocked</code> runs are always protected regardless of age.
<code>KNEO_SERV_RETENTION_IDEMPOTENCY_DAYS</code>	<code>unset</code>	Delete idempotency records by <code>updated_at</code> (they carry full response payloads and grow with traffic).
<code>KNEO_SERV_RETENTION_RUN_STATUSES</code>	<code>completed,failed,cancelled,timeout,expired</code>	Comma-separated terminal statuses eligible for the runs prune.
<code>KNEO_SERV_RETENTION_AUDIT_DAYS</code>	<code>unset</code>	Delete audit events older than this many days. The audit table grows unbounded otherwise — set this on long-lived deployments.
<code>KNEO_SERV_RETENTION_ARTIFACTS_DAYS</code>	<code>unset</code>	Delete old artifact files.

Variable	Default	Purpose
<code>KNEO_SERV_RETENTION_LOGS_DAYS</code>	unset	Delete old log files.
<code>KNEO_SERV_ARTIFACT_PATH</code>	unset	Root the artifacts retention pass prunes (<code>artifacts_days</code> no-ops without it).
<code>KNEO_SERV_LOG_PATH</code>	unset	Root the logs retention pass prunes (<code>logs_days</code> no-ops without it).
<code>KNEO_SERV_TRACE_MAX_EVENTS</code>	10000	Per-run trace buffer cap; 0 disables. Drops are counted and logged once.
<code>KNEO_SERV_MCP_CONNECT_TIMEOUT_SECONDS</code>	MCP per-attempt timeout, else 30	Bound on a lazy MCP session's connect; a hung server aborts instead of bricking the tool. Must be > 0. Connect timeouts cancel the connect coroutine cleanly, so configured MCP retries do retry them.
<code>KNEO_SERV_IDEMPOTENCY_LOCK_TTL_SECONDS</code>	300	Per-key idempotency lock TTL; raise it for deployments with long synchronous runs. Validated at startup.
<code>KNEO_SERV_PROVIDER_RETRY_ON_TIMEOUT</code>	false	Opt-in: retry a provider call after a per-attempt timeout. The timed-out attempt's thread is abandoned, not cancelled — only enable for idempotent call sites.
<code>KNEO_SERV_WORKFLOW_RETRY_ON_TIMEOUT</code>	false	Same opt-in for the workflow step/node retry surface (spec <code>max_retries + timeout_seconds</code>); restores the pre-0.9.0 retry-after-timeout behavior there.

Retention windows can also be set per-project in `.kneo/config.yaml` under a top-level `retention:` block — see [project_config.md § Retention](#). Env-var precedence: an env var, if set, overrides the project-config value for that field. Use the env-var path as the operator escape hatch when the host needs to deviate from per-project defaults.

Checkpoint storage

Variable	Default	Purpose
<code>KNEO_SERV_CHECKPOINT_COMPRESS_BYTES</code>	65536	Compress checkpoint payloads at or above this size.
<code>KNEO_SERV_CHECKPOINT_MAX_BYTES</code>	1048576	Hard checkpoint payload limit after compression.
<code>KNEO_SERV_CHECKPOINT_PREVIEW_CHARS</code>	1200	Preview length for oversized checkpoint values.
<code>KNEO_SERV_CHECKPOINT_MAX_LIST_ITEMS</code>	20	Maximum list items retained in previews.
<code>KNEO_SERV_CHECKPOINT_MAX_DICT_ITEMS</code>	50	Maximum dict items retained in previews.

Examples

Source: <docs/user/examples.md>

The repository ships a set of runnable specs and supporting Python helpers under [examples/](#). Use them to validate a local install, exercise the CLI, or as starting points for your own specs.

These specs are non-production placeholders — the `provider / model` fields point at common defaults and should be retargeted before any real use.

Run the commands below from the repository root. A spec resolves its tools by a dotted `implementation:` path (e.g. `examples.app_functions.web_search`), which is imported relative to your current working directory. The CLI puts the invocation directory on the import path for you, so `kneo spec compile examples/research_agent.yaml` works as written from the repo root — and a spec of your own resolves its `implementation:` modules relative to your project root the same way.

Feature → example matrix

Pick an example by the feature or surface you want to see. The *regression* examples double as the executable proof of a specific fix and run offline in CI (no provider keys, no network).

Feature / surface	Example	What it shows
Single agent + tools, env overlays	<code>research_agent.yaml (+ .dev / .staging / .prod)</code>	A base agent with function tools and per-environment overlays.
Graph workflow	<code>graph_review_workflow.yaml</code>	Conditional edges between steps.
Concurrent fan-out	<code>concurrent_review_workflow.yaml</code>	Parallel branches that join.
Group chat	<code>group_chat_workflow.yaml</code>	Multi-agent turn-taking.
Human-in-the-loop	<code>human_approval_workflow.yaml</code>	Pause at a human step, resume via the continuation API.
Human (smoke)	<code>smoke_human_workflow.yaml</code>	Minimal pause/resume on the <code>dummy</code> provider, for smoke tests.
Declarative MCP / agent-as-tool / workflow-as-agent	<code>declarative_spec.yaml</code>	The 0.8.0 declarative-parity features (compiles offline).

Feature / surface	Example	What it shows
Run-level timeouts	<code>run_with_timeout.py</code>	Deadlines + the <code>prune_timed_out_runs</code> sweep.
Project config / overlays	<code>project_config.yaml</code>	Per-environment overlays + policy + retention knobs.
Handoff <code>round_robin</code> (0.9.0 regression)	<code>handoff_workflow.yaml</code>	One turn per participant, then a <code>clean completed</code> .
Per-step <code>on_error</code> (0.9.0 regression)	<code>resilient_workflow.yaml</code>	<code>fallback</code> and <code>continue</code> policies actually executing.
MCP stdio transport (0.9.0 regression)	<code>mcp_stdio_workflow.yaml</code> + <code>mcp_stdio_server.py</code>	A real subprocess + stdio handshake on first tool call.
Nested workflow + human approval (0.10.0 HIGH #1)	<code>nested_workflow_human_approval.yaml</code>	A nested pipeline gated by a top-level human step; the human-inside-nested anti-pattern is rejected at validation.
Tool-stage guardrail enforcement (0.10.0 HIGH #2)	<code>guardrails_complete.yaml</code>	A declared tool-stage guardrail redacts PII; a workflow-stage one is rejected.
Secret redaction (0.10.0 MEDIUMs)	<code>redaction_demo.py</code>	Redaction across structured data, free text, and traces; pluralized credential keys redacted, usage keys preserved.
Custom middleware + adapter hop (0.10.0 MEDIUMs)	<code>custom_middleware_demo.py</code>	A custom middleware's <code>ToolResult.metadata</code> reaches the SDK; OTel context survives the worker-thread hop.

Spec files

[research_agent.yaml](#)

A single-agent research pipeline using a plan-act strategy with two tools (`web_search` , `webpage_reader`) and a sequential workflow that retrieves, analyzes, and summarizes.

```
kneo spec validate examples/research_agent.yaml
kneo spec compile examples/research_agent.yaml
kneo run --input "Analyze Nvidia AI business" --target workflow examples/research_agent.yaml
```

Three environment overlays show the overlay system in action:

- [research_agent.dev.yaml](#) — faster model, fewer iterations, tracing enabled.
- [research_agent.staging.yaml](#) — larger model, mid iterations, tracing enabled.
- [research_agent.prod.yaml](#) — conservative temperature, more iterations, step checkpointing, tracing.

```
kneo spec validate examples/research_agent.yaml --env prod
kneo spec bundle sign examples/research_agent.yaml \
  --output bundles/research_agent.prod.json --approved-by release-manager --env prod
```

[graph_review_workflow.yaml](#)

A graph workflow with conditional edges: retrieve → analyze → review → revise → finalize, where the review step routes to `revise` or `finalize` based on output. Demonstrates `GraphWorkflow`, conditional edges, and component agent references.

```
kneo spec compile examples/graph_review_workflow.yaml
```

[concurrent_review_workflow.yaml](#)

A concurrent workflow that fans out a single input to three reviewers (security, accessibility, performance) running in parallel. The platform collects each participant's response and returns the combined result. Demonstrates `ConcurrentWorkflow`, `participants:`-style declaration, and the fan-out / fan-in pattern.

```
kneo spec compile examples/concurrent_review_workflow.yaml
kneo run --input "Review the auth middleware refactor" \
  --target workflow examples/concurrent_review_workflow.yaml
```

[group_chat_workflow.yaml](#)

A group-chat workflow with three personas (proponent, skeptic, pragmatist) debating a design proposal over two rounds. Each round visits all participants in declaration order, so `rounds: 2` produces six total turns. Demonstrates `GroupChatWorkflow`, the `rounds:` knob, and ordered participant declaration for structured back-and-forth.

```
kneo spec compile examples/group_chat_workflow.yaml
kneo run --input "Should we adopt gRPC for service-to-service calls?" \
  --target workflow examples/group_chat_workflow.yaml
```

[human_approval_workflow.yaml](#)

Sequential workflow with a human-in-the-loop step (`kind: human`) between draft and publish. Use it to exercise the pause/resume API.

```
kneo run --input "hello" --target workflow --json examples/human_approval_workflow.yaml
# Output includes a continuation_id; resume with:
kneo human resume <continuation_id> --request-id <request_id> --approve
```

The deeper human-task documentation is in [design.md § 8.5](#) and the [HTTP API's /human-tasks/... endpoints](#).

Timeout branches

The `approval-reviewer` block in this spec declares a 24-hour timeout with `on_timeout: escalate` . Two other literals are available; the platform dispatches per [human_in_the_loop.md § 9](#) :

<code>on_timeout</code>	Lifecycle	Audit event(s)	Continuation
<code>fail</code> (default)	Run transitions to <code>expired</code>	<code>human.expired</code>	deleted
<code>continue</code>	Synthesizes an auto-approved <code>HumanResponse</code> and resumes the workflow	<code>human.continued</code> ; <code>human.continue_failed</code> on resume error	deleted on success or failure
<code>escalate</code>	Run stays <code>blocked</code> ; <code>escalated_at</code> stamped on the continuation; subsequent prune calls skip it (escalation fires once)	<code>human.escalated</code>	preserved (operator reassigns + resumes via the normal <code>/continuations/{id}/resume</code> path)

Auto-routing of an escalated task to a different reviewer is up to the operator's external workflow — the platform marks + audits the task as escalated, it does not auto-reassign. Operators call `PlatformManager.prune_expired_human_tasks()` (cron, scheduled run, manual sweep — same pattern as `prune_retention()` ; there is no built-in scheduler) to dispatch the timeout branch.

[run_with_timeout.py](#)

Worked walkthrough of the **run-level** timeout — distinct from the human-task timeout above.

`start_run_from_spec(..., timeout_seconds=N)` schedules a run with a wall-clock deadline written to `RunState.deadline_at`; `prune_timed_out_runs()` is the operator-callable sweep that force-cancels every `running` or `blocked` run past its deadline, transitions the state to `timed_out`, deletes any associated continuation, and emits a `run.timed_out` audit event.

```
python examples/run_with_timeout.py
```

Whichever timeout fires first wins. The dispatch matrix between run-level and human-task deadlines is documented in [human in the loop.md § 9](#) under *Run-level timeouts vs. human-task timeouts*.

[smoke_human_workflow.yaml](#)

Lightweight human-in-the-loop spec that uses the `dummy` provider so it runs without real provider credentials. Used by the deployment smoke script:

```
python scripts/deployment_smoke.py --base-url http://127.0.0.1:8000
```

See [deployment_smoke.md](#) for the full smoke sequence.

[declarative_spec.yaml](#)

The declarative-spec-parity features added in 0.8.0, in one spec:

- **MCP transport** — `mcp_servers` declares an `http` (or `stdio` / `sse`) server, and a tool binds to it with a `tool.mcp` block. The server config is built at compile time but only connected on first tool call, so it compiles offline. Enterprise mTLS fields (`verify` / `ca_bundle` / `client_cert` / `client_key_ref`) ride the same block.
- **Agent-as-tool** — `tool.agent` backs a tool with another component agent.
- **Workflow-as-agent** — `agent.as_agent` backs an agent with a workflow.

The build-order graph wires these cross-component references; a forward or cyclic reference is rejected at `validate`, not at runtime.

```
kneo spec validate examples/declarative_spec.yaml
kneo spec compile examples/declarative_spec.yaml
```

The skills surface is a runtime/API feature rather than a spec field. List the discoverable skills and overlay them per run against a running service:

```
# Read-only catalog (auth scope: specs:read)
curl -H "Authorization: Bearer $KEY" http://127.0.0.1:8000/v1/skills

# Per-request overlay on a run – add/disable within your scope; audited.
curl -X POST http://127.0.0.1:8000/v1/runs \
  -H "Authorization: Bearer $KEY" -H "Content-Type: application/json" \
  -d '{"input": "hi", "spec": {...}, "skills": {"disable": ["risky_skill"]}]}'
```

Project config

[project_config.yaml](#)

Reference `.kneo/config.yaml` content showing project metadata, service defaults, runtime defaults, and per-environment policy enforcement overlays. Copy into `.kneo/config.yaml` to bootstrap a new project, or use:

```
kneo config init --name research-agent-demo
kneo config show
```

The schema and overlay rules are in [project_config.md](#).

Helper Python

[app_functions.py](#)

Stub implementations for the tools and helpers referenced by `research_agent.yaml` (`compress_history`, `web_search`, `webpage_reader`, `summarize`). Ship-quality replacements would call real services; these just return formatted strings so the agent loop has something to do.

[human_functions.py](#)

Stub `draft_report` and `publish_report` used by the human-approval workflow. Same pattern as `app_functions.py`.

Adapting an example

1. Copy a spec into your project, e.g. `cp examples/research_agent.yaml my_agent.yaml`.
2. Replace the `model.provider` / `model.name` with your provider, and add the corresponding env-var reference under your project secrets.

3. Replace the tools with real ones — either Python functions registered through `ToolRegistry` or MCP servers (see [extending.md](#)).
4. Validate against your target environment: `bash kneo spec validate my_agent.yaml --env prod`
5. Compile to confirm the workflow builds: `bash kneo spec compile my_agent.yaml`
6. Run locally before deploying: `bash kneo run --input "<prompt>" --target workflow my_agent.yaml`

For deployment to a service, see [deployment.md](#).

0.9.0 additions: resilience, handoff, stdio MCP

Three examples landed with the 0.9.0 reliability cut, each doubling as the executable regression for a fixed surface (they run offline in CI — no provider keys, no network):

- **handoff_workflow.yaml** — a `round_robin` handoff: each participant takes one turn, then the run completes (`status: completed` , not the pre-0.9.0 `max_iterations` failure).

```
kneo run examples/handoff_workflow.yaml \ --input "triage this incident report" --target workflow --json
```

- **resilient_workflow.yaml** — per-step `on_error` : the `fetch` step always fails and falls back to `cached_fetch` ; `enrich` always fails and is skipped (`continue`). The final report shows `[cached]` content — both policies actually executing. Semantics: [run_lifecycle.md](#).

```
kneo run examples/resilient_workflow.yaml \ --input "quarterly metrics" --target workflow --json
```

- **mcp_stdio_workflow.yaml** — declarative MCP over stdio, backed by the bundled `mcp_stdio_server.py` (FastMCP, ships with the SDK's `mcp` dependency). The platform spawns the server as a subprocess on the tool's first call; the session lives on a dedicated event loop and is reused across calls. This is a **runtime transport proof** — the pre-0.9.0 stdio path failed on every invocation.

```
kneo run examples/mcp_stdio_workflow.yaml \ --input "count the words in this sentence" --target agent --json
```

The `retention: block` in `project_config.yaml` now also shows every retention knob, including 0.9.0's `idempotency_days`.

0.10.0 additions: regression-showcase examples

Four examples landed with the 0.10.0 cut, each the executable proof of a cluster-0 fix. They run offline in CI (no provider keys, no network). Where an agent would need a provider to drive a tool call, the

example proves enforcement at the seam instead (compile + wire + exercise, or a direct adapter call) — the YAML/script headers say which.

- **nested_workflow_human_approval.yaml** (*HIGH #1*) — a `kind: workflow` nested drafting pipeline gated by a **top-level** human-approval step. The run blocks at approval and publishes only the approved draft; burying the human step *inside* the nested workflow is rejected at validation (`E_STEP_WORKFLOW_NESTED_HUMAN`), so an approval gate can never be silently bypassed.

```
kneo run examples/nested_workflow_human_approval.yaml \ --input "the Q3 board report" --target workflow --json
```

- **guardrails_complete.yaml** (*HIGH #2*) — a `tool`-stage guardrail that redacts PII (an SSN pattern) from a tool result. Before the fix a `tool` / `workflow`-stage guardrail validated, satisfied the production `require_guardrails` gate, deployed, and was never enforced. As of **0.11.0** every tool-stage action is enforced (a *raising* action aborts the run via `GuardrailAbort`) and `workflow`-stage guardrails are enforced per step, so the 0.10.0 validator-rejects (`E_GUARDRAIL_ACTION_UNSUPPORTED` / `E_GUARDRAIL_STAGE_UNSUPPORTED`) no longer fire.
- **redaction_demo.py** — secret redaction across structured payloads, free text, and trace events. Pins the pluralized-key fix: `api_keys` / `refresh_tokens` are redacted while usage keys (`input_tokens` , `max_tokens`) survive.

```
python examples/redaction_demo.py
```

- **custom_middleware_demo.py** — a custom tool middleware plus the two adapter-hop fixes: a custom middleware's `ToolResult.metadata` reaches the SDK `ToolCallContext.metadata` , and the OpenTelemetry context survives the `run_awaitable_sync` worker-thread hop.

```
python examples/custom_middleware_demo.py
```

Project `.kneo/` config

Source: docs/user/project_config.md

Each project keeps a `.kneo/` directory for local service configuration, generated artifacts, and logs. This page covers the contents and the overlay/policy story; for the runtime variables read from the environment (rather than from project config), see <environment.md>.

```
.kneo/
  config.yaml
  README.md
  artifacts/.gitkeep
  logs/.gitkeep
```

The `.kneo/config.yaml` demonstrates:

- default project name and owner
- service URL
- local state/artifact/log paths
- default spec
- environment overlays
- runtime defaults
- model defaults
- policy defaults
- environment-variable secret references
- retention windows (per-project)

Environment-specific policy enforcement can be configured under `environments`.

`<name>.policy_enforcement`:

```
environments:
  dev:
    policy_enforcement:
      enabled: false
  staging:
    policy_enforcement:
      require_tool_permissions: true
      blocked_diagnostic_codes: [E_UNSAFE_TOOL_IMPORT, E_UNSAFE_FUNCTION_IMPORT]
  prod:
    policy_enforcement:
      require_tool_permissions: true
      deny_unrestricted_tools: true
      require_human_review: true
```

```
require_guardrails: true
blocked_diagnostic_codes: [E_UNSAFE_TOOL_IMPORT, E_UNSAFE_FUNCTION_IMPORT]
```

Policy enforcement runs after spec overlays and project defaults are applied. `kneo spec validate --env prod`, `kneo spec compile --env prod`, `kneo spec policy-report --env prod`, and `kneo run --env prod` all honor the resolved environment policy.

Retention

Retention windows for runs, checkpoints, queue records, continuations, audit events, artifacts, and log files live in a top-level `retention:` block. Each field is a count of days to keep; unset fields disable pruning for that category. Values must be zero or greater.

```
retention:
  runs_days: 30
  checkpoints_days: 14
  queue_days: 7
  continuations_days: 21
  audit_days: 45
  artifacts_days: 60
  logs_days: 90
```

`audit_days` prunes the audit-event log purely by age (it reaps events with no associated run too); the audit table grows unbounded otherwise, so it is the table most likely to exhaust disk on a long-lived deployment.

The same seven retention fields can be overridden per-host via env vars (`KNEO_SERV_RETENTION_RUNS_DAYS` and friends; see [environment.md § Retention](#)). **Precedence is env var > project config > unset.** Set the project-config field for the per-project default; set the env var to deviate on a specific host (staging vs. prod, etc.) without editing the committed `.kneo/config.yaml`.

The retention values feed

```
kneo_serv.maintenance.retention.RetentionPolicy.from_project_and_env(config.retention),
```

which the operator can pass to `PlatformManager.prune_retention(policy=...)` on whatever cadence makes sense for the deployment (cron, scheduled workflow, manual operator action).

Local / self-hosted LLM endpoints

The native (`openai`) runtime can target any OpenAI-compatible server — **Ollama**, **vLLM**, **llama.cpp**, **LocalAI** — by setting `base_url` (and, if the server requires one, an API key) under the agent spec's `model.extra`:

```

agent:
  name: on-prem-assistant
  runtime_preferences:
    preferred_mode: native
    native_provider: openai
  model:
    provider: openai
    name: llama3          # the model the local server serves
  extra:
    base_url: http://localhost:11434/v1  # e.g. Ollama
    api_key_ref: LOCAL_LLM_KEY          # resolved from the env (see below)

```

- **base_url** — the OpenAI-compatible endpoint. Omit it to use the hosted OpenAI API as before; this field is purely additive.
- **api_key_ref** — the *name of a secret reference*, resolved at runtime through the `SecretResolver` (it reads the project `extra_env` map, falling back to an env var of the same name). Prefer this over a literal so the key is never baked into a persisted or signed spec bundle.
- **api_key** — a literal-key escape hatch for throwaway/local use. It is a [sensitive key](#), so it is redacted from audit events and list responses — but it still lives in the stored spec, so `api_key_ref` is the recommended path.

If neither key field is set, the runtime falls back to the provider's normal env var (`OPENAI_API_KEY`), which is fine for a keyless local server.